



1
XIAMEN
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DIGITAL LOGIC

Assignment of Chapter Three (B)

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A. FILL-IN-BLANK

2

1. When designing multiple output combinational logic circuits, we need to fully consider the sharing terms/ common terms , so that the circuits can be further simplified.
2. The relationships of a combinational logic circuit can be represented by truth table, and logic function expression , such two methods.
3. The methods to determining whether a circuit can have hazards include Algebraic method, and Karnaugh map method .
4. When designing a combinational logic circuit with “don’t care” terms, we must make the full use of the arbitrariness of the “don’t care” terms. This statement is (True).
5. The race phenomenon is caused by incorrect circuit designs. This statement is (False).
6. In combinatorial logic circuits, a hazard caused by race is a transient error that does not cause the circuit’s a steady state result. This statement is _____ (True).

B. CIRCUIT ANALYSIS

3

$$F_1 = AB + A\bar{C} + \bar{C}D$$

- No variable has its true and complement forms simultaneously; therefore, cannot have the race and hazard.

- $F_2 = AB + \bar{A}CD + BC$
- It has A and $\bar{A}$; thus, it will have the race situation.
- BCD = 000, $F_2 = 0$
- BCD = 001, $F_2 = 0$
- BCD = 010, $F_2 = 0$
- BCD = 011, $F_2 = \bar{A}$
- BCD = 100, $F_2 = A$
- BCD = 101, $F_2 = A$
- BCD = 110, $F_2 = A+1 = 1$
- BCD = 111, $F_2 = A + \bar{A} + 1 = 1$
- Therefore, it will not have hazard.

- $F_3 = (A + \bar{B})(\bar{A} + \bar{C})$
- It has A and $\bar{A}$; thus, it will have the race situation.
- **CANNOT transform its expression.**
- BC=00, $F_3 = 1$
- BC=01, $F_3 = \bar{A}$
- BC=10, $F_3 = A$
- BC=11, $F_3 = A + \bar{A}$
- Therefore, it will have hazard.

C. CIRCUIT DESIGN

6

1. Use the NAND gates to implement a combinational logic circuit, whose input is a one-bit 2421 code number. The circuit is to determine whether the input is a prime number; if so, $F=1$, if not $F=0$.
2. Step 1: list its truth table:

$$F(A, B, C, D) = \sum m(2, 3, 11, 13) + \sum d(5 \sim 10)$$

3. Steps 2 and 3; list its logic function and its transformation:

$$F(A, B, C, D) = \overline{B}C + A\overline{C}D = \overline{\overline{\overline{B}C} \cdot \overline{A\overline{C}D}}$$

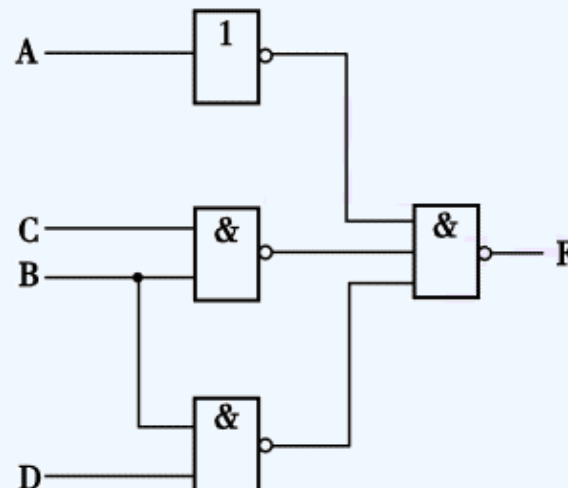
Decimal number	8421code	2421code	Excess-3code
0	0000	0000	0011
1	0001	0001	0100
2	0010	0010	0101
3	0011	0011	0110
4	0100	0100	0111
5	0101	1011	1000
6	0110	1100	1001
7	0111	1101	1010
8	1000	1110	1011
9	1001	1111	1100

2. Use the NAND gates to design a rounding function circuit, whose input is a one-bit 8421 code number. If the number is larger than and equal to 5, $F=1$, if not $F=0$.

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	d
		⋮		⋮
1	1	1	1	d

$$F(A,B,C,D) = \sum m(5 \sim 9) + \sum d(10 \sim 15)$$

$$F(A,B,C,D) = A + BC + BD = \overline{\overline{A}} \overline{\overline{BC}} \overline{\overline{BD}}$$



3. Design a “FULL ADDER/ SUBTRACER”, it has a input M, which control the circuit’s addition and subtraction processes. If M=0, the circuit is a FULL ADDER; if M=1, the circuit is a FULL SUBTACER.

- (Tips: the circuit has 4 inputs, M, A, B, C; A denotes the summand/ minuend; B denotes addend/minuend; C denotes for the carry/borrow from the lower bit; F denotes the sum/difference; G denotes the carry/borrow to the high bit)

M	A	B	C	F	G	M	A	B	C	F	G
0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	1	1	0	1	0	0	1	1	1
0	0	1	0	1	0	1	0	1	0	1	1
0	0	1	1	0	1	1	0	1	1	0	1
0	1	0	0	1	0	1	1	0	0	1	0
0	1	0	1	0	1	1	1	0	1	0	0
0	1	1	0	0	1	1	1	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1

$$M=0: F(A, B, C) = \sum m(1, 2, 4, 7)$$

$$G(A, B, C) = \sum m(3, 5, 6, 7)$$

$$M=1: F(A, B, C) = \sum m(1, 2, 4, 7)$$

$$G(A, B, C) = \sum m(1, 2, 3, 7)$$

$$M=0: F = A \oplus B \oplus C;$$

$$G = AB + AC + BC = \overline{\overline{A} \cdot \overline{B} \cdot \overline{C}}$$

$$M=1: F = A \oplus B \oplus C;$$

$$G = \overline{A}B + \overline{A}C + BC = \overline{\overline{\overline{A} \cdot \overline{B} \cdot \overline{C}}}$$

